

Table 5.4 Estimated MI Parameters

Scenario	a_1	a_2	a_3	a_4	b_1
All Data	708	0.55	0.71	0.50	0.98
SE	100	0.03	0.02	0.05	0.04
R^2	0.42				

Next we performed our non-linear least squares regression for the full model without holding any parameter value fixed. We allowed the model to determine the set of parameters using the specified constraints to ensure feasible values. The results of the regression are shown in [Table 5.4](#). The table includes parameter standard errors from the non-linear regression and has a non-linear $R^2 = 0.42$. These statistical results indicate a strong fit. Additionally, notice these results are all within the ranges we previously determined. This provides a higher degree of confidence in our results.

The best fit equation using these estimated parameters is:

$$I = 708 \cdot \text{Size}^{0.55} \cdot \sigma^{0.71} \quad (5.16)$$

$$MI = 0.98 \cdot I \cdot \text{POV}^{0.50} + (1 - 0.98) \cdot I \quad (5.17)$$

To further evaluate costs and determine differences across categories we further categorized the data into samples that consisted of large and small cap companies, buy and sell orders, and a breakdown by market cap and order size. In all cases, there was a high fit of the data. The non-linear R^2 ranged from 0.40 to 0.43. These results are shown in [Table 5.5](#).

An analysis of the costs estimates over time did not find any differences across buy and sell orders when holding volatility and trading strategy constant. However, in practice, managers and traders often find sell orders to be more costly than buy orders for various reasons. First, buy orders are cancelled more often than sell orders. As the price moves too high the advantage and incremental alpha decreases and managers are better suited investing in an alternative stock. There are substitution stocks for buy orders but not for sell orders. Once a stock has fallen out of favor and the manager decides to remove the stock from the portfolio they will complete the order regardless of the price. Therefore, managers do not always realize the entire cost of the buy order because they rarely factor in opportunity cost. But the entire cost of the sell order is always realized. Second, managers typically sell stocks at a more aggressive rate